

DC Current Control for a Single-Stage Current Source Inverter in Motor Drive Application

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Abstract—The current source inverter (CSI) is a power electronics topology that allows for the realization of variable speed drives (VSD). Compared to the most common voltage source inverter (VSI), which can be directly connected to a voltage source, the CSI needs a prestage to generate a constant current bus. This article therefore seeks to challenge this “accepted” consideration that a CSI always needs this precircuit and seeks to remove this circuit by proposing an innovative i_{dc} current control scheme. The proposed scheme is applied to a single stage motor drive driven by a CSI converter. It is shown how implementing this control scheme removes the need for the front-end stage, thus removing an unnecessary converter and optimizing the efficiency at the same time. The CSI state-space equations are presented and the developed models are verified using simulations. Stability analysis of small-signal model is considered through Nyquist criterion with the robustness in presence of variations of the most important system parameters. Experimental results driving a permanent magnet synchronous machine (PMSM) are shown confirming the validity of the proposed control, potentially paving the way to a larger adoption of the CSI topologies for motor drive applications.

Index Terms—Current source inverter, dc current control, electric drive system, electric machine control.

I. INTRODUCTION

THE importance of controlled electric drives (systems that include a power electronic (PE) converter, an electrical machine (EM), and a controller) is well known through all

sectors of the industry. They facilitate cost effective and efficient renewable energy generation, enable the adoption of more electric aircraft (MEA) technology [1], [2], and provide traction power for electric propulsion in railways [3], ships [4], and cars [5].

Modern, variable speed drives allow for excellent performance in terms of system efficiency, controllability, and overall performance. This is of course partially due to the advent of advanced PE converters of the last three decades or so, aided by significant advances in computational technologies.

Known as the workhorse of the industry, the most widespread PE converter for low and medium power (from hundreds of W to hundreds of kW) applications is of course the three-phase full-bridge converter. This simple and robust structure has made the converter the obvious choice for a lot of applications and industries.

In most applications, the inverter stage of the full converter, usually includes a voltage source inverter (VSI) [6]. The advances in the last decades in terms of the development of power semiconductors (IGBT or SiC MOSFETs with no reverse blocking capability) has consolidated the role of the VSI as the most common type of inverter used in converter applications (see [7] and [8]).

A well-known but very rarely implemented alternative to the VSI is the current source inverter (CSI). An authoritative comparison between CSIs and VSIs is given in [9], while [10], [11] study the use of CSIs in photovoltaic applications. The CSI is made up of six bidirectional blocking devices in a full-bridge configuration. Whereas the VSI modulates the dc link voltage to the output, the CSI modulates the dc link current, with the obvious advantage of a simple current control, short circuit protection and absence of electrolytic capacitor to form the dc link. On the other hand, the CSI requires a front-end converter to implement a constant current dc link. This, combined with the need for bidirectional blocking power devices are the drawback of this topology. Due to these important aspects, the CSI has traditionally been relegated to a secondary role in terms of VSDs and their control.

Recently, however, the need for higher performance systems, especially in terms of power density, efficiency, reliability, and robustness is projecting the field of VSDs into a different scenario. The industrial pull represented by initiatives like the MEA [12] and the MEE [5], in their quest toward lighter and greener systems, are demanding ever higher power density

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performances [13]. Therefore a lot of research effort is today being focused on higher operational speeds.

Indeed, there are a number of applications, related to transport electrification, where the operating cycle consists of steady-state operation with maximum power and maximum speed, and the voltage boosting capability of the CSI is demonstrated to allow for a strong reduction of weight of the machine [14].

Therefore, it is on these considerations that the CSI's role as the second stage of a full-bridge converter is perceived to be advantageous. In this context, the well-known drawback of CSIs, namely, the input inductor, is immediately mitigated since higher speeds mean higher frequencies, as these automatically result in smaller magnetic components. For a VSI, the higher frequencies only result in minimal variations on the input dc capacitance, which is mostly sized to withstand the current ripple. The other main advantage of CSIs is the intrinsic output current control, that automatically removes the associated risks of insulation failure related to high speed voltage control such as high dV/dts . Therefore, it can be safely stated that for some modern applications and operational conditions, the CSI can be an excellent solution that compares excellently against VSIs.

Having said that, it is however still true that even for higher frequency drives, the CSI still requires a conditioning prestage. It is therefore this particular aspect that this article seeks to address: The possibility of removing the prestage, for application which works constantly at higher operating frequency. If the need for a front-end converter was to be removed, then it is perceived that a much wider acceptance of the CSI in the field of VSDs can be achieved. This article seeks to achieve this by implementing a new control scheme of the input current, that can replace that need for the converter.

Historically, various solutions aiming to control the input current i_{dc} have been proposed in literature, with most of them revolving around the concept of cancellation and addressing of the harmonic content of the input current. The classical solution for an induction machine (IM) drive is proposed in [15] which comprises two inverter stages, where the first one is a current source rectifier (CSR) that actively controls the i_{dc} current and the second stage is a six-device PWM-CSI. Another methodology for the implementation of a CSI drive is proposed in [16]. The front-end of the CSI is implemented with a buck converter for the dc input voltage. In [17], two different solutions of CSI with front-end converter were proposed. A buck-boost converter and a single-ended primary inductance converter (SEPIC) develop two extended single-stage dc-ac inverters with bidirectional power conversion ability. The solution proposed in [18] uses a dc-dc front-end with two additional devices and proposes a two-stage control strategy which guarantees system performance maximization, within the whole operating range. The basic "message" that can be concluded from all these works is that the conditioning prestage for a CSI is essential to allow the output regulation of the converter. A constant i_{dc} current must be guaranteed, for the PWM modulation to work properly. Considering all the above, then it can be concluded that this can be achieved by adopting a control scheme that allows the CSI to operate in boost mode for significant periods of the operation cycle. The purpose of the article is therefore to propose and

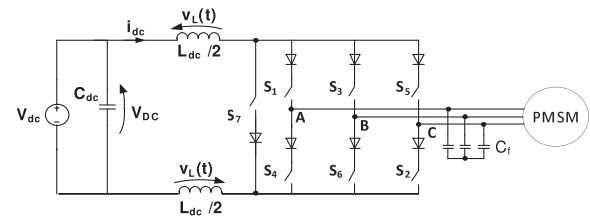


Fig. 1. Schematic of CSI7.

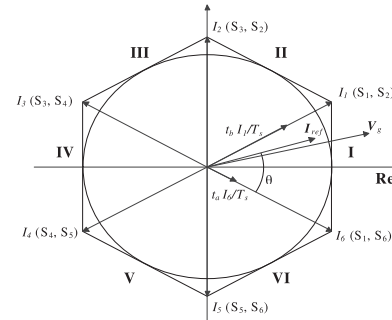
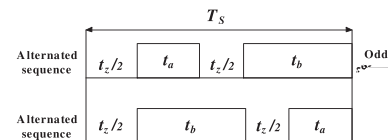


Fig. 2. Space vector modulation.



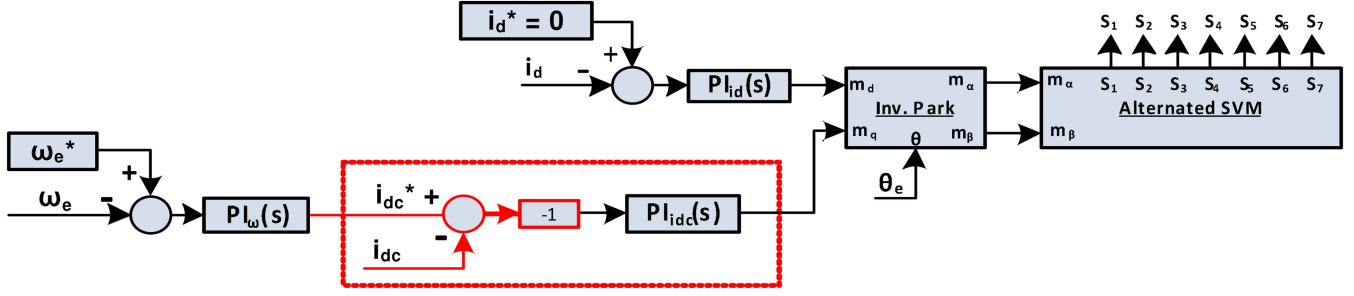


Fig. 4. Proposed control strategy with i_{dc} current control of a PMSM without front-end converter stage.

The working principle of the CSI assumes that a constant current source is available. Common passive diodes configurations or other setups that are commonly found in industry cannot guarantee an appropriate current signal. In fact, this constant current has always been a defining challenge of the CSI.

The CSI operates by chopping the dc current into the output terminals through a PWM scheme. Several modulation strategies have been proposed in literature [20], [21]. The initial step is the definition of the modulation index, which is described by (1), where $|i_{ref}|$ is the module of the current space vector and i_{dc} is the dc input current

$$m = \frac{|i_{ref}|}{i_{dc}}. \quad (1)$$

In Fig. 2, the maximum value of m is $\frac{\sqrt{3}}{2}$, defined by the inner circle of the hexagon and therefore the range of m can be described by

$$0 < m < \frac{\sqrt{3}}{2}. \quad (2)$$

The principle of the SVM is the capability to choose how to synthesize the reference output vector by selecting the best combination of active and zero vectors. During the transition between two vectors, an overlapping time where two vectors are applied simultaneously is introduced to prevent the open-load condition that would create an over voltage due to the inductive dc link. This is equivalent to the dead-time for a VSI. To enhance the system operation, it was decided to use an optimized SVM, known as the alternated space vector modulation, as proposed in [11]. Using this scheme results in better current quality. Fig. 3 shows the commutation strategy of the ASVM scheme. In this AVSM, the values of the active states A/B are given by t_a and t_b described by $d_a T_s$ and $d_b T_s$, respectively. The time t_z of zero state is computed as

$$t_z = T_s - (t_a + t_b). \quad (3)$$

The normalized time intervals of the two active states d_a and d_b are

$$\begin{cases} d_a = \frac{2}{\sqrt{3}} m \sin\left(\frac{\pi}{6} - \theta\right) \\ d_b = \frac{2}{\sqrt{3}} m \sin\left(\frac{\pi}{6} + \theta\right). \end{cases} \quad (4)$$

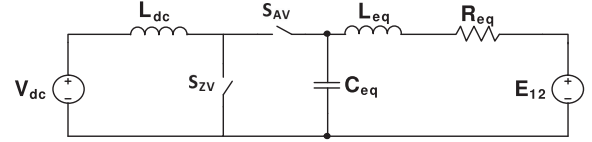


Fig. 5. Equivalent circuit of CSI converter during the application of the zero vector and one of the active vectors.

III. PROPOSED CONTROL STRATEGY

In this section, the proposed control strategy is introduced. For this work, the structure of the platform being considered is similar to the scheme given in Fig. 1. The control strategy is based on classical field oriented control (FOC), where the novelty is highlighted in the red box in Fig. 4.

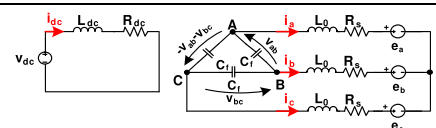
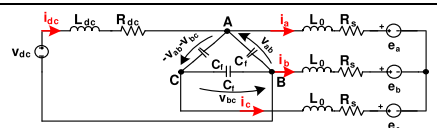
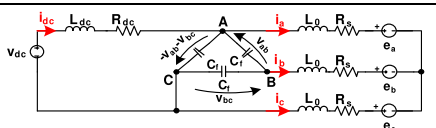
In most applications requiring VSDs, a stable bus current is not available. However, in transport applications, dc voltage distribution systems are almost always available [22], [23]. Considering all this, then the proposed control scheme can be developed as follows. First, m is defined as $\sqrt{m_d^2 + m_q^2}$, where m_d and m_q values represent the dq motor current components normalized with respect to the dc link current. After an inverse Park transformation, the normalized m_α and m_β currents components are computed. Thus, the ASVM for a CSI is used to determine the interval times of the required active and zero vectors with the calculation of the modulation index m .

This results in a control scheme that removes the need of a front-end converter stage. This is shown in Fig. 4 which supplies the required i_{dc} current. As can be observed, the whole concept is based on the CSI working in boost operation.

Regulation takes place through a new closed loop control on the i_{dc} current. Considering the similarity between a CSI's behavior and a synchronous-boost converter, such as the one shown in Fig. 5, then it can be said that the L_{eq} , R_{eq} , and E_{12} represent the equivalent circuit of the machine during the application of one of the active vectors. The zero vector is implemented with S_{ZV} and the active vector with S_{AV} . In fact, the higher output current/power can be transferred with higher values of t_z , i.e., the duty cycle of this equivalent boost converter. For this reason, in Fig. 4, there is the inversion of the i_{dc} current loop sign.

Equation (1) shows the relation between i_{dc} current and output current $|i_{ref}|$ under the hypothesis that i_{dc} current is in continuous conduction mode (CCM). The CCM condition is

TABLE I
STATE SPACE EQUATIONS OF THE CSI FOR ONE SECTOR UNDER CCM CONDITION

$z : 0 \leq t < t_z$	$a : t_z \leq t < t_z + t_a$	$b : t_z + t_a \leq t < T_s$
		
$\dot{x}_z = A_z x + B_z u$	$\dot{x}_a = A_a x + B_a u$	$\dot{x}_b = A_b x + B_b u$
$\begin{bmatrix} -\frac{R_{dc}}{L_{dc}} & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$		

B. Small-Signal Model

In this Section, the small-signal model of the three-phase CSI connected to a PMSM is derived.

In (6) v_{dc} represents the system input for constant values of m and ω , but in practice, m is used to regulate the output of the current source inverter. This results in the large signal model (6) to be linearized around a steady-state operating point by defining small perturbations as

$$\begin{aligned} x_{dq} &= X_{dq} + \delta x_{dq} \\ v_{dc} &= V_{dc} + \delta v_{dc} \\ m &= M + \delta m \\ \omega_e &= \Omega_e + \delta \omega_e. \end{aligned} \quad (7)$$

Also, the back EMF terms e_d and e_q are substituted by

$$\begin{aligned} e_d &= L_0 \omega_e i_q \\ e_q &= -L_0 \omega_e i_d - \omega_e \lambda_{PM}. \end{aligned} \quad (8)$$

The resulting linearized small-signal model is therefore as given by

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \delta i_{dc} \\ \delta v_d \\ \delta v_q \\ \delta i_d \\ \delta i_q \end{bmatrix} &= \underbrace{\begin{bmatrix} -\frac{R_{dc}}{L_{dc}} & -\frac{\sqrt{3}}{2L_{dc}}M & -\frac{1}{2L_{dc}}M & 0 & 0 \\ \frac{\sqrt{3}}{3C_f}M & 0 & \Omega_e & -\frac{1}{2C_f} & \frac{\sqrt{3}}{6C_f} \\ \frac{1}{3C_f}M & -\Omega_e & 0 & -\frac{\sqrt{3}}{6C_f} & -\frac{1}{2C_f} \\ 0 & \frac{1}{2L_0} & \frac{\sqrt{3}}{6L_0} & -\frac{R_s}{L_0} & \Omega_e \\ 0 & -\frac{\sqrt{3}}{6L_0} & \frac{1}{2L_0} & -\Omega_e & -\frac{R_s}{L_0} \end{bmatrix}}_{A_{sm}} \underbrace{\begin{bmatrix} \delta i_{dc} \\ \delta v_d \\ \delta v_q \\ \delta i_d \\ \delta i_q \end{bmatrix}}_x \\ &+ \underbrace{\begin{bmatrix} \frac{1}{L_{dc}} & -\frac{\sqrt{3}}{2L_{dc}}V_d - \frac{1}{2L_{dc}}V_q & 0 \\ 0 & \frac{\sqrt{3}}{3C_f}I_{dc} & V_q \\ 0 & \frac{1}{3C_f}I_{dc} & -V_d \\ 0 & 0 & I_q \\ 0 & 0 & -I_d - \frac{\lambda_{PM}}{L_0} \end{bmatrix}}_{B_{sm}} \underbrace{\begin{bmatrix} \delta v_{dc} \\ \delta m \\ \delta \omega_e \end{bmatrix}}_u. \end{aligned} \quad (9)$$

The closed loop control that regulates the error between the reference current i_{dc}^* and the measured i_{dc} with a proportional-integral controller which assure the regulation of the error (see Fig. 6), as presented above represent one of the main innovation of this article. The stability proof is made by computing the transfer function $G(s)$ between the input δm and the output δi_{dc} considering the matrices A_{sm} and B_{sm} in (9) and the output matrix C_{sm} as $[1, 0, 0, 0, 0]$ to select δi_{dc} , while the input-output matrix D_{sm} is equal to zero because there is no direct connection between the input and the output.

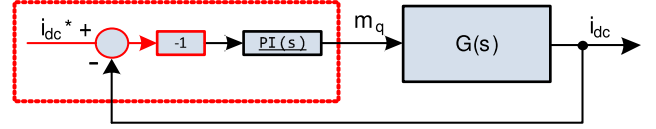


Fig. 6. Control scheme considered for the stability of the equilibrium point.

The open loop function $L(s)$ given by $PI(s)G(s)$ presents poles in the left half plane, i.e., $\Re\{\lambda_i\} < 0$. This ensures the stability of the closed loop, as it is necessary that the Nyquist diagram of $L(s)$ presents no turns around the point -1 . For the stability analysis, the Nyquist criterion is preferred to a Bode diagram because the system passes the 0-dB axe multiple times, so the stability is not regular but conditional. All eigenvalues of the closed loop system are located in the left-half s -plane, indicating an asymptotically stable system. But this does not mean that the system is stable in presence of parameters change, so variation of the eigenvalues must be studied. As shown in Fig. 7, the system remains stable in case of perturbations on the parameters R_s , L_0 , C_f , m , and ω_e .

C. Overall System Stability

The stability of the other loops, i.e., the i_d current and of the speed is here studied. Supposing the the i_d current loop is perfectly decoupled, the open loop transfer function $G_{ol}^{i_d}(s)$ choosing for the regulation a PI-controller is given by

$$G_{ol}^{i_d}(s) = \frac{K_p^{i_d}s + K_i^{i_d}}{s} \frac{1}{sL_0 + R_s} \quad (10)$$

obtaining the closed loop transfer function

$$G_{cl}^{i_d}(s) = \frac{K_p^{i_d}s + K_i^{i_d}}{L_0s^2 + (K_p^{i_d} + R_s)s + K_i^{i_d}} \quad (11)$$

the closed loop stability is ensured choosing proper values of $K_p^{i_d}$, $K_i^{i_d} > 0$.

Regarding the stability of the speed loop, the dynamic of the i_{dc} current loop can be considered as a low-pass filter

$$G_{cl}^{i_{dc}}(s) = \frac{i_{dc}(s)}{i_{dc}^*(s)} = \frac{\omega_{cc}}{s + \omega_{cc}} \quad (12)$$

where ω_{cc} is the current loop bandwidth which is higher than the speed loop bandwidth. The open loop transfer function $G_{ol}^\omega(s)$ of the speed loop can be computed as

$$G_{ol}^\omega(s) = \frac{K_p^\omega s + K_i^\omega}{s} \frac{1}{Js} \frac{\omega_{cc}}{s + \omega_{cc}} \quad (13)$$

where J is the inertia of the motor. The stability of the closed loop transfer function $G_{cl}^\omega(s)$ given by

$$G_{cl}^\omega(s) = \frac{(K_p^\omega s + K_i^\omega)\omega_{cc}}{Js^3 + J\omega_{cc}s^2 + K_p^\omega\omega_{cc}s + K_i^\omega\omega_{cc}} \quad (14)$$

is assessed by Routh criterion respecting the conditions $K_p^\omega\omega_{cc} > K_i^\omega$ and $K_i^\omega\omega_{cc} > 0$.

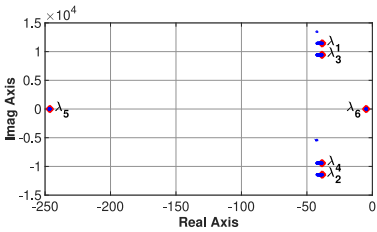
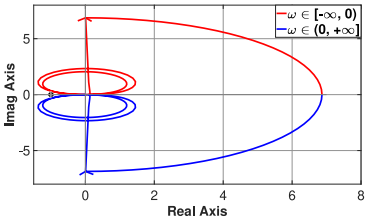


TABLE III
CURRENT SOURCE INVERTER SIMULATION PARAMETERS

Name	Value	Unit
v_{dc}	270	V
i_{dcmax}	140	A
m_{qmax}	0.86	-
ω_{ref}	5150	rad/s
L_{dc}	2	mH
$T_{S-AltPWM}$	50	μs
T_{ov}	1	μs
C	6.6	μF

TABLE IV
PMSMs ELECTRICAL PARAMETERS

Parameter	Value
Phase current (rms)	114.3 A
R_s	0.05 Ω
L_σ	rad
	Ω
Mechanical power	32.2 k

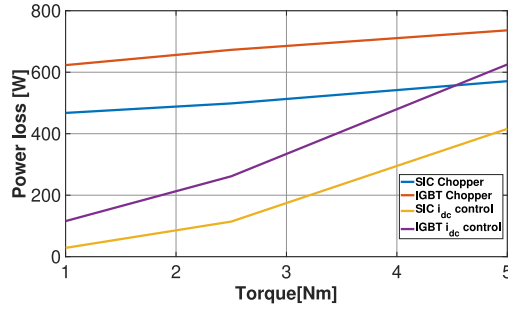


Fig. 14. Total power losses comparison between i_{dc} control and chopper.

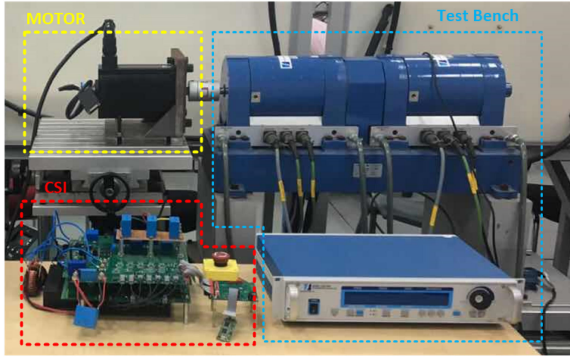


Fig. 15. CSI7 inverter prototype with SiC devices and test bench.

Thus two sets of results are achieved from both the traditional and the proposed configurations. The results are compared in Fig. 14.

The choice between these two technologies are a consequence of the 1200-V voltage rating. In fact, 1200-V silicon MOSFETs, with an acceptable ON-state resistance, are today still not commercially available. If a lower voltage rating was acceptable, the solution of Si MOSFET optimized for conduction losses (i.e., CoolMos technology without fast recovery body diode) could also be considered. In fact, no reverse recovery of the MOSFET body diode happens in the CSI topology. The configuration with i_{dc} current control and SiC devices has the lowest power losses. It should be noted, that the dc chopper would allow to operate the machine even when the back EMF is lower than the dc input voltage. The comparison of Fig. 14, therefore, is valid only when the operating speed of the machine is high enough so that the back EMF exceeds the input voltage.

VI. EXPERIMENTAL RESULTS

To validate the proposed control scheme and thus prove that a CSI converter can be used without a preconditioning stage, a complete prototype of the CSI7 has been realized. The prototype system has been developed based on SiC devices, as used in the simulation section (Wolfspeed C2M0025120D). A commercial TMS320F2812 digital signal controller that implements the modulation strategy and the control with a sampling and switching frequency of 10 kHz was used.

The PE is connected to a PMSM machine and a loading system, as shown in Fig. 15. A machine on derated power and with limited speed is used, and an input voltage of 28 V as the

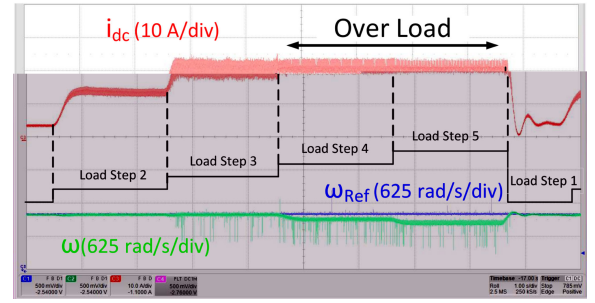


Fig. 16. Experimental results: current control in normal and over-load condition. The time division is 1 s/div.

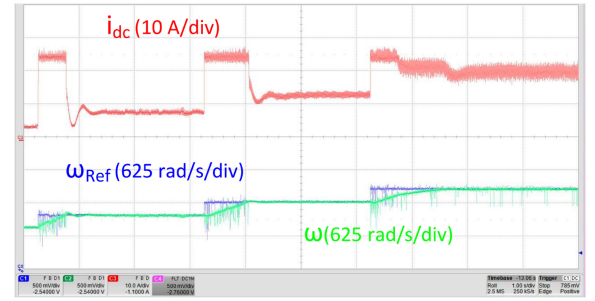


Fig. 17. Experimental results: current control with different speed references. The time division is 1 s/div.

dc bus is selected. A digital to analog converter (DAC) is used to plot the internal variables of the digital signal processor (DSP) onto an oscilloscope.

Having setup the prototype and its test bench, a series of experimental runs were designed aimed to prove the concept of the proposed scheme and also investigate its stability when various operating conditions are considered. The results from these tests are then compared against those achieved from the simulation model, where the same operating conditions as those in Section V are recreated experimentally.

Fig. 16 shows the relevant waveform, when the CSI is driving the test machine at a constant speed. In this case, a load profile similar to Section V is generated with the load system. The matching to Fig. 11 is excellent. It can be observed that for an overload condition, the speed loop cannot track the reference anymore and the measured speed decreases following the mechanical dynamics. Spikes in the DAC channels are caused by disturbances in the measurement due to the switching devices and not by glitches of the control.

Fig. 17 repeats the same condition of Fig. 13 where a speed profile is generated. The maximum speed is limited to 2500 r/min due to the bearings of the test machine. It is possible to see that the control is adapting the input current to match the load condition. The maximum input current limits the acceleration, causing the speed to increase linearly.

Fig. 18 shows the three phase currents i_u , i_v , and i_w with the dc link current i_{dc} in steady-state condition.

The distortion in the output current waveform in Fig. 18 is caused by the alternate SVM [11] in this critical condition: Low values of i_{dc} and high modulation indices. The ASVM introduce a third harmonic of the fundamental motor frequency,

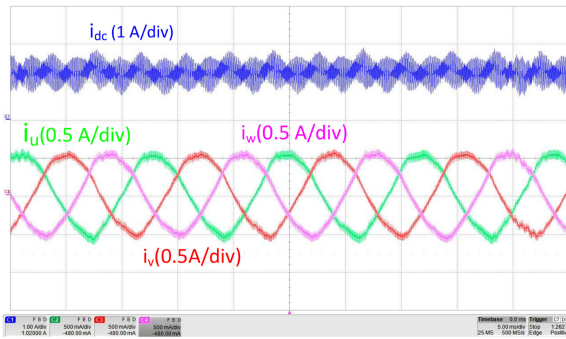


Fig. 18. Experimental results: three phase currents and i_{dc} in steady state. The time division is 5 ms/div.

consequently distorting the output current. If a base SVM is used the absence of low component frequency is ensured. Also using the alternate SVM with lower modulation index would make the current distortion not noticeable. This is due to the proposed control scheme that does not need the ac output currents measures. The drawback of this solution is the absence of any disturbance rejection from i_{dc} variation that can determines an harmonic distortion in the output currents if the i_{dc} waveform presents harmonics of the fundamental motor frequency.

VII. CONCLUSION

In this article, a novel control scheme has been proposed allowing to remove the need for the conditioning circuit at the input of a CSI for all the applications in which the electric motor has to work for almost all the time at high speed, i.e., pumps, compressors. This approach has been theoretically analyzed, simulated, and experimentally validated. To investigate the performance of the CSI drive used here, a stability analysis of the CSI was conducted by deriving the average state-space model of the converter with the proof of the stability by Nyquist criterion and eigenvalues position in presence of perturbation on the most important parameters of the model. All the simulation results and proof-of-concept experiments confirm that the CSI drive can operate as a single-state solution with intrinsic voltage boost capability. Having removed this drawback, then the inherent advantages of a CSI drive can be full exploited: Boost capability, absence of electrolytic capacitors, and low motor insulation stress due to the low dV/dt . In addition, in this work it has also been shown how a CSI without the front-end converter can be implemented with the aim of driving a PMSM motor with low power losses thanks to the reduced number of stages and the load-dependent control. The main perceived limitation of the proposed scheme is the intrinsic boost operation, which limits the minimum speed.

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